GGG-S-65a
APRIL 5, 1961
SUPERSEDING
Fed. Spec. GGG–S–65
December 8, 1954 and
GGG–S–61a
November 3, 1942, IN PART

FEDERAL SPECIFICATION

SAWS, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers manually-operated hand saws for cutting wood, metal, plastics and like materials. The saw consists of a flat blade with teeth on one or two edges, a handle and attaching screws. Frame-type saws and blades, and one- and two-man crosscut saws are not included.

1.1.1 *Federal specification coverage.*—This specification covers only the types and classes of hand saws generally purchased by the Federal Government and is not intended to include all of the types and styles which are commercially available.

1.2 Classification.

1.2.1 *Types, styles and classes.*—Hand saws shall be of the following types, styles, and classes as specified (see 6.1) :

Type I—Back.
Type II—Cabinet.
Type III—Cable.
Type IV—Compass.
Type V—Crosscut.
  Style A—Skew back.
  Style B—Straight back.
    Class 1—Full taper ground.
    Class 2—Flat ground.
Type VI—Docking.
Type VII—Dovetail.
Type VIII—Keyhole.
  Style A—Woodcutting.
  Style B—Metal-cutting.
Type IX—Metal-cutting.
Type X—Miter box.
Type XI—Nested.
Type XII—Plumbers'.
Type XIII—Pruning.
  Style A—Single edge.
  Style B—Double edge.
Type XIV—Rip.
  Style A—Skew back.
  Style B—Straight back.
    Class 1—Full taper ground.
    Class 2—Flat ground.
Type XV—Adjustable handle.
  Class 1—Plastic handle.
  Class 2—Metal handle.
  Style A—Compass blade.
  Style B—Keyhole blade.
  Style C—Nail-saw blade.

1.2.2 *Sizes.*—Hand saws shall be of the sizes shown in tables I through XV for each type, style, and class as specified (see 3.2, 3.6 and 6.1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invita-

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tion for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of saws otherwise conforming to this specification.

3.2 Material.

3.2.1 *Blades.*—Saw blades shall be made of hot-rolled saw steel which can be heat-treated and tempered to the hardness requirements hereinafter specified. The steel shall be free from cracks, segregations, inclusions, seams, hammer marks, high spots or other imperfections that may affect serviceability. The steel shall be cross rolled and show a fine fracture grain.

3.2.2 *Handles.*

3.2.2.1 *Wooden handles.*—Wooden handles shall be made of thoroughly seasoned (not over 12-percent moisture) straight-grained, clear hardwood, free from any imperfections which may affect serviceability or durability.

3.2.2.2 *Metal handles.*—Metal handles shall be of corrosion-resisting metal or shall have a suitable rust-resisting finish.

3.2.2.3 *Plastic handles.* — Plastic handles shall be of a suitable fire-resistant or self-extinguishing plastic (see 4.5.4).

3.2.3 Screws, medallions, nuts and rivets, unless otherwise specified in the contract or order, shall be made of metal showing a Rockwell B hardness of not less than 60 and shall have a suitable rust-resisting coating or shall be of corrosion-resisting metal.

3.3 Nominal length.—The nominal length of all saws, except type X, shall be the distance between the teeth nearest the ends of the saw blade. The thickness of the cutting edge shall be measured next to but not including the teeth.

3.4 Spring set teeth.—Saws with spring set teeth shall have every tooth the same degree; alternate teeth shall be set to the same side of the saw. Ample clearance shall be provided on each side of the blade of all saws to prevent binding under service conditions.

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3.5 Points and teeth per inch.—Points and teeth per inch shall be measured as shown on figure 1 (see 6.1). (Note.—Always one more point than teeth per inch.)

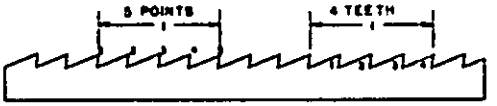


FIGURE 1.—Points and teeth per inch.

3.6 Tolerances.—On dimensions specified hereinafter as approximate, a plus or minus tolerance of 10 percent will be permitted; a plus or minus tolerance of 0.003 inch will be permitted when uniform thickness is specified; and a plus or minus one point per inch will be permitted in points per inch, except on type V and type X.

3.7 Thickness.—Unless otherwise specified hereinafter, the blades shall be of uniform thickness throughout.

3.8 Finish.

3.8.1 *Wood handles.*—Wood handles shall have polished and rounded edges. All exposed surfaces including the inside of the handgrip shall be finished smooth, stained and either wax-coated or lacquered with a durable weather -resistant lacquer or varnish.

3.8.2 *Metal and plastic handles.* — Metal and plastic handles shall have smooth grips and be free from sharp edges and projections.

3.9 Marking for identification.—Saws shall be permanently marked with the manufacturer's name, or with a trademark of such known character that the source of manufacture may be readily determined. The marking shall be applied in a manner that will preclude damage to the saws.

3.9.1 *Special marking.*—Special marking, when specified in the contract or order, shall be permanently applied in accordance with the specifications listed herein. Special marking shall not damage or render the saw defective.

3.10 Type I, back saw.—Type I saws shall be securely held in a reinforcing steel back approximately ¾ inch wide and 0.109 inch thick, extending the full length of the back of the blade. The blade shall be hardened and tempered to a Rockwell C hardness of not less than 48 nor more than 52. The teeth shall be spring set, bevel-filed and adapted for both crosscutting and ripping. The handle shall be of hardwood and secured to the blade by means of three screws and nuts. The saws shall be similar to figure 2 and shall conform to the dimensions shown in table I for the length specified (see 6.1).

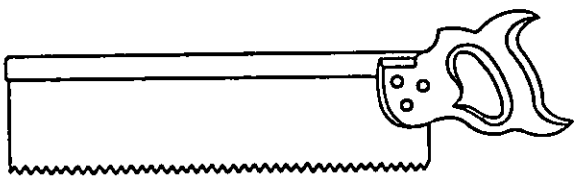


FIGURE 2.—Type I, back saw.

TABLE I.—*Type I, back saws*

Length of cutting edge	Width under back		Points per inch	Thickness of blade	
	Minimum	Maximum		Minimum	Maximum
Inches	*Inches*	*Inches*		*Inch*	*Inch*
12	2¾	3¼	14	0.028	0.038
16	3⅛	3⅞	12	.028	.042

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3.11 Type II, cabinet saw.—Type II saws shall consist of a flat saw blade with two parallel edges, toothed on both edges. One edge shall be toothed for ripping, the other for crosscutting. The blade shall be hardened and tempered to a Rockwell C hardness of not less than 46 nor more than 49. The handle shall be of hardwood and secured to the blade by a steel bolt and wing nut with rust-resisting finish. Cabinet saws shall be similar to figure 3 and shall conform to the dimensions shown in table II.

3.12 Type III, cable saw.—Type III saws shall consist of a flat tapered saw blade toothed on two edges for cutting cable. The handle shall be of hardwood and shall be secured by not less than three screws and nuts. The blade shall be heat-treated and tempered to a Rockwell C hardness of not less than 50 nor more than 58. Cable saws shall be similar to figure 4 and shall conform to the dimensions shown in table III for the length specified (see 6.1).

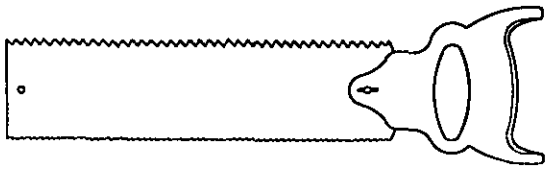


FIGURE 3.—Type II, cabinet saw.

TABLE II.—*Type II, cabinet saws*

Length of cutting edge (approx.)	Width of blade (approx.)	Points per inch		Thickness of blade	
		Rip side	Crosscut side	Minimum	Maximum
Inches	*Inches*			*Inch*	*Inch*
12	3⅛	10	14	0.032	0.042

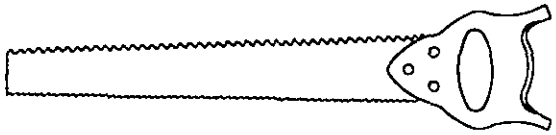


FIGURE 4.—Type III, cable saw.

TABLE III.—*Type III, cable saws*

Length of cutting edge (approx.)	Width of blade		Points per inch	Thickness of blade	
	Point (approx.)	Handle (approx.)		Minimum	Maximum
Inches	*Inches*	*Inches*		*Inch*	*Inch*
12	2	2¾	8 and 15	0.042	0.058
14	2⅛	2¾	8 and 15	.042	.058
16	⅞	2⅞	10 and 13	.042	.058

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3.13 Type IV, compass saw.—Type IV saws shall be of uniform thickness throughout the length of the cutting edge. The handle end of the back of the blade may be of the same thickness as the cutting edge, in which case there shall be a graduated taper from the handle to the point; or the back may be of uniform thickness throughout, in which case the back shall be tapered with reference to the cutting edge. The blade shall be hardened and tempered to a Rockwell C hardness of not less than 44 nor more than 49. The teeth shall be spring set, bevel filed, and adapted for both crosscutting and ripping. The handle shall be made of thoroughly seasoned, straight-grain hardwood, and shall be secured to the blade by means of two screws and nuts. Saws shall be similar to figure 5 and shall conform to the dimensions shown in table IV for the length specified (see 6.1).

3.14 Type V, crosscut saw.—The crosscut saws shall be capable of withstanding the bending test of 4.4.3.

3.14.1 *Style A, skew back.*—Type V, style A saw blades shall be of the skew-back type, true and full taper ground, that is, uniform in thickness along the entire length of the cutting edge, uniformly tapered in thickness from the tooth edge to the back and along the back from the handle to the point. The tapers shall be uniform and not less than 0.013 inch. In no case shall the point of the back edge be less than 0.032 inch nor more than 0.044 inch, and the thickness of the back at the handle shall be not less than the thickness of the cutting edge. The teeth shall be bevel filed spring set which shall not extend more than half the depth of the tooth, and adapted for cross-cutting. The blades shall be hardened and tempered to a Rockwell C hardness of not less than 46 nor more than 52; and tensioned. The handles shall be made of hardwood. The handles on 26-inch saws shall be secured to the blade by at least 4 screws or 3 screws and one medallion. The handles on saw sizes smaller than 26 inch shall be secured by at least 3 screws or 2 screws and one medallion. Type V, style A saws shall be similar to figure 6 and conform to the dimensions shown in table V for the length specified (see 6.1). The handles shall be firmly secured to the blade so as not to rock or show other signs of looseness under actual or simulated service conditions. The handles shall have a handhold opening of not less than 1¼ by 3½ inches measured at the largest portions of the opening.

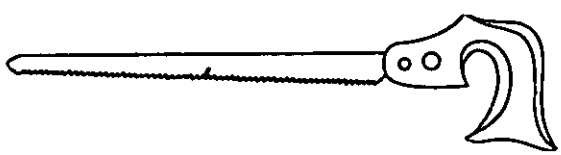


FIGURE 5.—Type IV, compass saw.

TABLE IV.—*Type IV, compass saws*

Length of cutting edge	Points per inch	Thickness of cutting edge		Thickness of back at point		Blade width at handle	
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Inches		*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*
10	9	0.049	0.065	0.035	0.049	1	1¼
12	9	.049	.065	.035	.049	1	1¼
14	9	.049	.065	.035	.049	1	1⅜

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FIGURE 6.—Type V crosscut, style A.

TABLE V.— *Type V, crosscut saws*

Length of cutting edge	Points per inch	Width	
		Butt (approx.)	Point (min.)
Inches		*Inches*	Inches
20	10-11	5	1¼
22	8-10-11	5	1¼
24	8-9-10-11	6	1¼
26	7-8-9-10-11-12	6⅛ (6 min.)	1⅛

3.14.2 *Style B, straight back.* — Type V, style B hand saws shall conform to type V, style A (see 3.14.1) except that as to the shape of the back which shall be straight, and the handle of the 24- and 26-inch saws shall be secured by 4 or more screws or a medallion and 3 or more screws. The handle of the 20- and 22-inch saws shall be secured by three or more screws or a medallion and two or more screws. The handles shall have a handhold opening of not less than 1¼ by 3½ inches, measured at the largest portions of the opening. Saws shall be similar to figure 7 and shall conform to the dimensions shown in table V for the length specified (see 6.1).

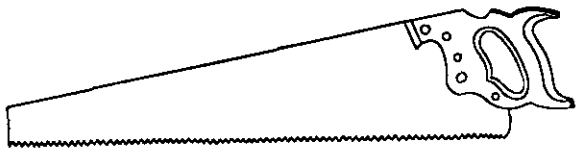


FIGURE 7.—Type V, crosscut, style B.

3.14.2.1 *Class 2, flat ground.* — Type V, style B, class 2 saws shall be similar to type V, style B, class 1, except for the blade which shall be ground to a uniform thickness throughout, or tapered from the cutting edge to the back of the blade. The cutting edge of the blade shall be of uniform thickness throughout its length, and the thickness shall be not less than 0.032 inch nor more than 0.042 inch. The handle shall be secured by three or more screws or a medallion and two or more screws.

3.15 Type VI docking saw.—Type VI saws shall be of the skew-back type. The blades shall be of uniform thickness throughout the length of the cutting edge. They shall be taper ground from the handle end of the saw to the end of the blade and from the cutting edge to the back. They shall be hardened and tempered to a Rockwell C hardness of not less than 48 nor more than 52. They shall be filed and set. The handle shall be of metal with corrosion-resisting finish and shall be secured to the blade by two or more rivets. Docking saws shall be similar to figure 8 and shall conform to the dimensions shown in table VI for the length specified (see 6.1).

3.16 Type VII, dovetail saw.—Type VII saws shall be fitted with a reinforcing steel back extending the full length of the blade. The blade shall be hardened and tempered to a Rockwell C hardness of not less than 46 nor more than 49. The teeth shall be spring set, bevel filed, and adapted for both crosscutting and ripping. A handle of the straight grip type shall be securely fastened to reinforcing back. Dovetail saws shall be similar to figure 9 and shall conform to the dimensions shown in table VII for the length specified (see 6.1).

3.17 Type VIII, keyhole saw.

3.17.1 *Style A, woodcutting.*—Type VIII, style A, woodcutting keyhole saws shall be of uniform thickness throughout the length of the cutting edge. The blade may be either of uniform thickness throughout or tapered, at the option of the supplier. The blade shall be hardened and tempered to a Rockwell C hardness of not less than 44 nor more than 49. The teeth shall be spring set, bevel filed, and adapted for both cross-cutting and ripping. The handle shall be made of thoroughly seasoned, straight-grain hardwood, and shall

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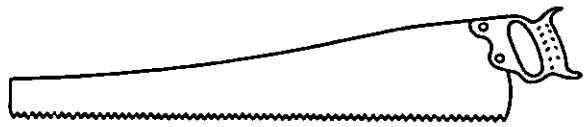


FIGURE 8.—Type VI, docking saw.

TABLE VI.—*Type VI, docking saws*

Length of cutting edge	Points per inch	Thickness of cutting edge		Thickness of back at point		Width	
		Minimum	Maximum	Minimum	Maximum	Handle end (approx.)	Point (approx.)
Inches		*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*
24	4½	0.045	0.055	0.032	0.042	6¼	2½
30	4½	.045	.055	.032	.042	6½	2½

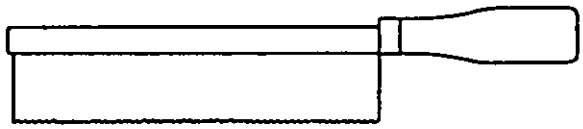


FIGURE 9.—Type VII, dovetail saw.

TABLE VII.—*Type VII, dovetail saws*

Length of blade	Width under back	Points per inch	Thickness of blade	
			Minimum	Maximum
Inches	*Inches*		*Inch*	*Inch*
8	1⅝	16	0.028	0.035
10	1⅝	16	.028	.035
12	1⅝	17	.028	.035

be secured to the blade by means of two screws and nuts. Saws shall be similar to figure 10 and shall conform to the dimensions shown in table VIII.

3.17.2 *Style B, metal-cutting.*—Type VIII, style B, metal-cutting keyhole saws shall consist of a detachable metal handle and a short tapered metal-cutting blade. The handle shall be of the pistol-grip type and shall be designed to fit a ⅝-inch wide steel saw blade and hold it securely in place. The blade width shall be tapered from ⅝ inch at the handle end to $\frac{5}{16}$ inch at the point. The blade shall have a flexible back and hardened teeth capable of resisting the cutting action of a file,

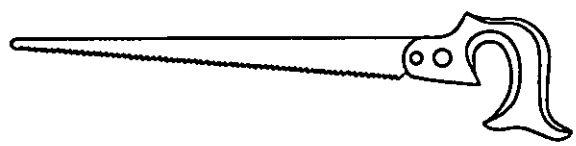


FIGURE 10.—Type VIII, style A, keyhole saw, woodcutting.

TABLE VIII.—*Type VIII, style A, keyhole saws, woodcutting*

Length of cutting edge	Points per inch	Thickness of cutting edge		Blade width at handle	
		Minimum	Maximum	Minimum	Maximum
Inches		*Inch*	*Inch*	*Inch*	*Inch*
10	10	0.042	0.058	⅛	$\frac{13}{16}$

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and shall be capable of withstanding the metal cutting test of 4.4.2.2. The saw shall be similar to figure 11 and shall conform to the dimensions shown in table IX for the length specified (see 6.1).

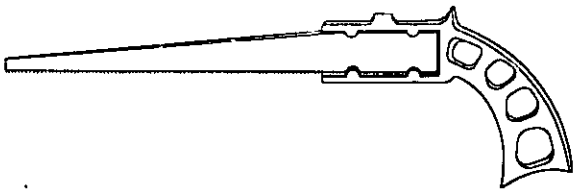


FIGURE 11.—Type VIII, style B, keyhole saw, metal-cutting.

TABLE IX.—*Type VIII, style B, keyhole saws, metal-cutting*

Length of cutting edge	Points per inch	Thickness of blade		Width at heel
		Minimum	Maximum	
Inches		*Inch*	*Inch*	*Inch*
5½	18	0.025	0.040	¾
7½	24	.025	.040	¾
8	18	.046	.061	¾

3.18 Type IX, metal-cutting. — Type IX saws shall be of the straight-back type and either taper-ground or taper and hollow ground. They shall be of uniform thickness throughout the length of the cutting edge. The back of the saw may be of uniform gage throughout or tapered from the handle to the point. The blades shall be hardened and tempered to a Rockwell C hardness of not less than 52 nor more than 58, and shall be capable of withstanding the test of 4.4.2.1. The teeth shall not be set, but shall be milled straight across. The handle shall be made of hardwood and shall be secured to the blade by means of either three or four screws and nuts. Type IX saws shall be similar to figure 7 and shall conform to the dimensions shown in table X for the length specified (see 6.1).

3.19 Type X, miter box saw.—Type X saws shall be securely held in a reinforcing steel back approximately 1 inch wide, 0.109 inch thick and extending the full length of the back. The blade shall be hardened and tempered to a Rockwell C hardness of not less than 48 nor more than 52. The teeth shall be spring set, bevel filed and adapted for both crosscutting and ripping. The handle shall be of hardwood and secured to the blade by not less than three screws and nuts. Unless otherwise specified (see 6.1), they shall be similar to either figure 12 (a) or 12 (b), and shall conform to the dimensions shown in table XI. Points per inch shall be as specified (see 6.1).

3.20 Type XI, nested saw.—Type XI nested saws shall consist essentially of three blades and a detachable hardwood handle similar to figure 13, provided with a device for quickly and securely attaching handle to blade. The nest shall consist of a keyhole saw, compass saw, and nail-cutting saw. The handle

TABLE X.—*Type IX, metal-cutting saws*

Length of cutting edge	Points per inch	Thickness of cutting edge		Thickness of back at handle		Thickness of back at point		Width	
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Butt (approx.)	Point (max.)
Inches		*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*
18	15	0.042	0.049	0.032	0.049	0.022	0.025	5¼	1⁹⁄₁₆
20	15	.042	.049	.032	.049	.022	.025	5½	1⁹⁄₁₆
22	15	.042	.049	.032	.049	.022	.025	5½	1⁹⁄₁₆
24	15	.042	.049	.032	.049	.022	.025	5½	1⁹⁄₁₆

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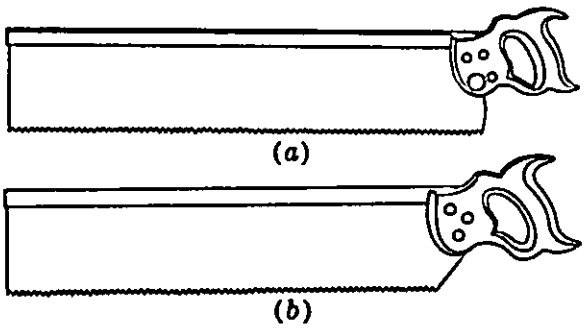


FIGURE 12.—Type X, miter box saws.

TABLE XI.—*Type X, miter box saws*

Full length of blade	Width under back (approx.)	Points per inch	Thickness	
		Minimum	Minimum	Maximum
Inches	*Inches*		*Inch*	*Inch*
26 ------	4	11	0.037	0.043

end of each blade shall be properly designed to fit the handle; otherwise, the keyhole saw blade shall conform to the 10-inch type VIII keyhole saw (see 3.17), and the compass saw blade to the 12- or 14-inch type IV compass saw (see 3.13). The nail-cutting saw blade shall be hardened and tempered to a Rockwell C hardness of not less than 49 nor more than 52. The nail cutting blade shall be capable of withstanding the metal cutting test of 4.4.2.2. The blade shall be of uniform thickness throughout the length of the cutting edge. The handle end of the back of the blade may be of the same thickness as the cutting edge, in which case there shall be a graduated taper from the handle to the point or the back may be of uniform thickness throughout, in which case the back shall be tapered with reference to the cutting edge. The teeth shall be set and milled straight across. They shall conform to the dimensions shown in table XII.

3.21 Type XII, plumbers' saw.—Type XII saws shall be of a straight-back type, toothed on one edge and have a reversible hardwood handle secured by one bolt and wingnut. It shall be suitable for cutting studs, rafters, and flooring, where nails are imbedded, as well as soil and gas pipe. The blade shall be of uniform thickness on the cutting edge and

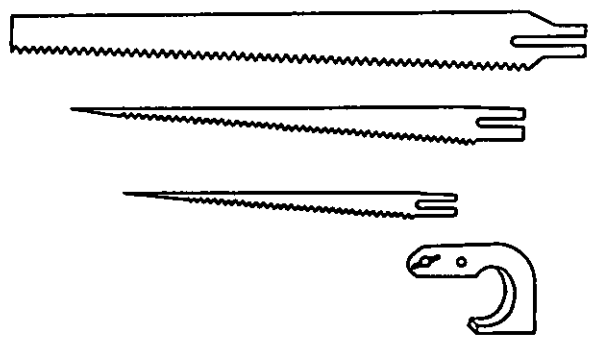


FIGURE 13.—Type XI, nested saws (nailcutting, keyhole and compass blades with handle.

TABLE XII.—*Type XI, nail-cutting blade for nested saws*

Length of cutting edge	Points per inch	Thickness of cutting edge		Thickness of back at point		Blade width at handle	
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Inches		*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*
16 to 18 -----	15	0.042	0.048	0.032	0.035	1⅛	2⅛

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taper ground for clearance. It shall have a Rockwell C hardness of not less than 48 nor more than 52. The teeth shall not be set but shall be milled straight across. Type XII saws shall be similar to figure 14 and shall conform to the dimensions shown in table XIII.

3.22 Type XIII, pruning saws.

3.22.1 *Style A, single edge.* — Style A, single-edge pruning saws shall have a straight back and a hardwood handle, which may be laminated. The blade shall be hardened and tempered to a Rockwell C hardness of not less than 48 nor more than 52. The blade shall be filed and set and adapted for cutting large or small limbs. The handle shall be large enough to readily permit the use of gloves, and shall be attached to the blade by at least three screws. Type XIII, style A pruning saws shall be similar to figure 15 and shall conform to the dimensions shown in table XIV.

3.22.2 *Style B, double edge.* — Style B, double-edge pruning saws shall have a straight tapering blade with a double horn hardwood handle secured by three screws and nuts. The blade shall be approximately 18 inches long with teeth on both edges. The

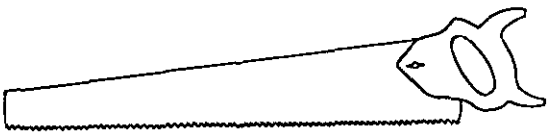


FIGURE 14.—Plumbers' saw.

TABLE XIII.—*Type XII, plumbers' saw*

Length of cutting edge	Points per inch	Thickness of cutting edge		Thickness of back		Width at point (approx.)	Width at butt (approx.)
		Minimum	Maximum	Minimum	Maximum		
Inches		*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*
18	12	0.035	0.049	0.022	0.028	1¾	3⅛

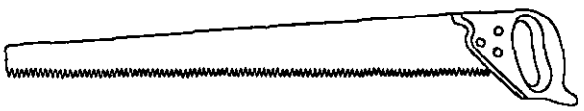


FIGURE 15.—Type XIII, style A, pruning saw.

TABLE XIV.—*Type XIII, style A, pruning saws, single edge*

Length of cutting edge	Points per inch	Thickness on cutting edge		Width at point		Width at handle end	
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Inches		*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*
26	4⅛	0.042	0.056	1⅛	2	3	3⅛

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teeth on one edge shall be bevel filed and set for crosscutting with eight points per inch, and the other edge shall have teeth known as lumberman style. The blades shall be hardened and tempered to a Rockwell C hardness of not less than 48 nor more than 52. Type XIII, style B pruning saws shall be similar to figure 16.

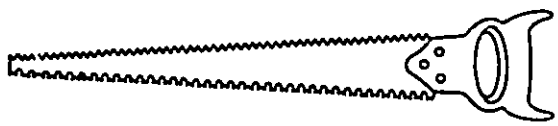


FIGURE 16.—Type XIII, style B, pruning saw.

3.23 Type XIV, rip saws.—The rip saws shall be capable of withstanding the bending test of 4.4.3.

3.23.1 *Style A, skew back.*—Style A, hand rip saws shall conform to type V, style A crosscut saws, (see 3.14.1) except as to dimensions, and that the teeth shall be filed straight across and adapted for ripping. Type XIV, style A saws shall be similar to figure 6 and shall conform to the dimensions shown in table XV.

TABLE XV.—*Type XIV, styles A and B, rip saw, skew back and straight back*

Length of cutting edge	Points per inch	Width	
		Butt (approx.)	Point (approx.)
Inches		*Inches*	*Inches*
26	5½	6½ (6 min.)	1¾

3.23.2 *Style B, straight back.*

3.23.2.1 *Class 1, full taper ground.*—Class 1 rip saws shall conform to type V, style B, class 1 crosscut saws (see 3.14.2.1) except as to dimensions and that the teeth shall be filed straight across and adapted for ripping. Saws shall be similar to figure 7 and shall conform to the dimensions shown in table XV.

3.23.2.2 *Class 2, flat ground.*—Type XIV, style B, class 2 saws shall be similar to type V, style B, class 1, saws (see 3.14.2.2), except for the blade which shall be ground to a uniform thickness throughout or tapered from the cutting edge to the back of the blade. The cutting edge of the blade shall be of uniform thickness throughout its length, and the thickness shall be not less than 0.032 inch nor more than 0.042 inch. The handle shall be secured by three or more screws or a medallion and two or more screws.

3.24 Type XV—adjustable handle saw.

3.24.1 *Type XV handles.*—The blade-holding device of both the class 1 and 2 handles shall be so designed as to hold either of the blade ends shown in figure 19 securely, without any blade movement in any direction.

3.24.2 *Class 1, plastic handle.* — Class 1 handles shall be of a pistol-grip design with a swiveling chuck and locking pin which permits the blade to be rotated through 360 degrees and can be securely locked in at least eight positions at equal intervals of 45 degrees. The chuck shall be of such design as to accommodate styles A, B and C blades. The finished surfaces of the clutch, locking pin and knurled nut shall be made of noncorrosive metal, or shall have a corrosive-resistant finish. The handle shall pass the flammability test of 4.5.4 and shall be similar to figure 17.

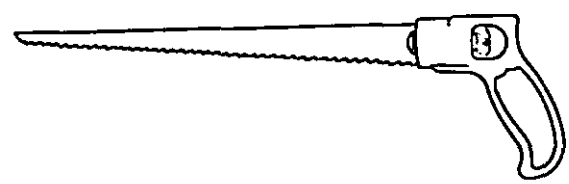


FIGURE 17.—Type XV, class 1, plastic handle.

3.24.3 *Class 2, metal handle.* — Class 2 handles shall be similar to class 1, except that it shall be cast metal. The cast metal part shall be lacquered or painted to resist corrosion. The class 2 handle shall be similar to figure 18.

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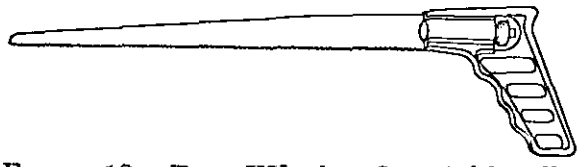


FIGURE 18.—Type XV, class 2, metal handle.

3.24.4 *Styles A, B and C blades.*—Styles A, B and C blades shall be of uniform thickness throughout the length of the cutting edge and back.

3.24.5 *Handle end of blades.*—The handle end of the style A, B and C blades shall conform to either of the two ends shown in figure 19, and must be interchangeable between the type XV, class 1 and class 2 handles.

3.24.6 *Style A compass blade.*—Style A adjustable handle saw blades shall have a Rockwell "C" hardness value of from 44 to 49. The teeth shall be formed, set and filed. The blade shall have 7 teeth (8 points) per inch.

3.24.7 *Style B, keyhole blade.*—Style B adjustable handle saw blade shall have a Rockwell "C" hardness value of from 58 to 62 and shall contain 24 teeth (25 points) per inch. The blades shall be capable of withstanding the cutting test specified in 4.5.2.1.

3.24.8 *Style C, nail saw blade.*—Style C adjustable handle saw blade shall have a Rockwell "C" hardness value of from 58 to 62 and shall contain 10 teeth (11 points) per inch. The blades shall be capable of withstanding the metal cutting test specified in 4.5.2.2.

3.25 **Workmanship.**—The blades of the saws shall be true and straight throughout their length and width. The workmanship, finish, and appearance of saws shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Lot.**—For the purpose of sampling, a lot shall consist of all saws of the same type, style, class and size offered for delivery at one time.

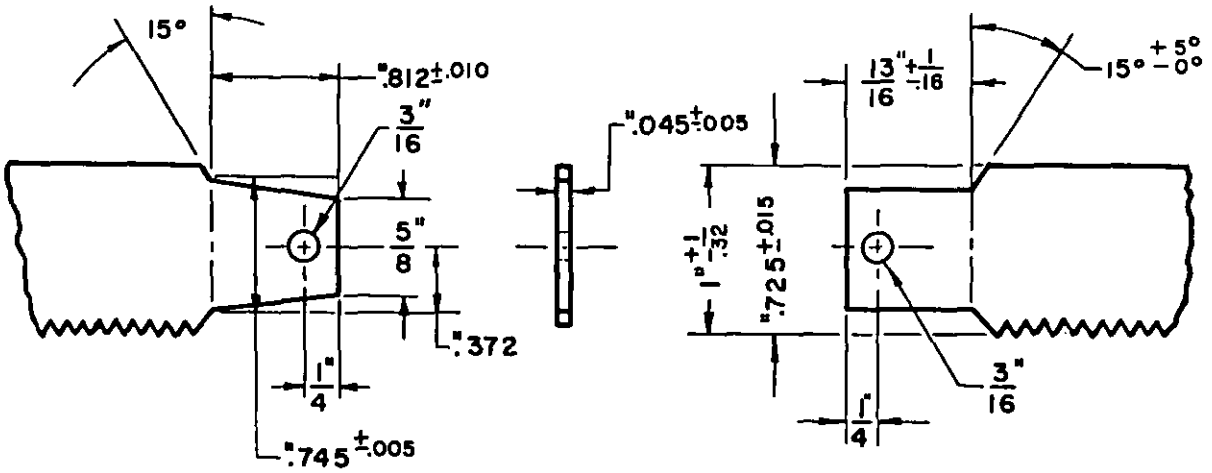


FIGURE 19.—Optional handle ends for type XV styles.

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4.3 Sampling for acceptance inspection.

4.3.1 *Sampling for examination and hardness test.*—A random sample of saws shall be taken in accordance with table XVI from each inspection lot of material offered for inspection, with lot acceptance based on the following sampling inspection requirements in accordance with Military Standard MIL–STD–105. Any sample saw which does not conform to this specification shall be rejected, and if the number of nonconforming saws in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE XVI.—*Sampling for examination and hardness test (AQL (approx.) equals 1.5 percent defective)*

Number of saws in the lot	Number of saws in the sample
40 and under	Inspection level III
41 to 800	Inspection level II
801 and over	Inspection level I

Note.—The inspector may institute tightened or reduced inspection in accordance with section 9 of MIL–STD–105.

4.3.2 *Sampling for metal cutting and bending tests.*—A random sample of saws shall be selected from each inspection lot in accordance with Military Standard MIL–STD–105 at inspection level L–8. The acceptable quality level shall be 2.5 percent defective. However, each of the sample sizes indicated in the table in the appendix shall be used and the acceptance number shall be zero until the acceptable quality level permits one or greater. Each of the sample saws shall be subjected to the applicable test of 4.4. Any sample saw which does not meet the requirements shall be rejected, and if the number of nonconforming saws in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.3 *Sampling for flammability test (type XV, class 1).* — A random sample of saws shall be selected from each inspection lot in accordance with the appendix to Military Standard MIL–STD–105 at inspection level L–5. Each sample shall be subjected to the flammability test of 4.4.4. If any sample saw fails the test, the lot shall be rejected.

4.4 Examination. — Each sample saw selected in accordance with 4.3.1 shall be carefully examined to determine compliance with requirements such as material, workmanship, finish, marking, and dimensions. Examination shall be conducted as specified in table XVII.

TABLE XVII.—*Classification of defects in accordance with Military Standard MIL–STD–105*

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type, size, length, style and (if applicable) number of points per inch of saw not as specified.
102	Blade defective; evidence of cracks, segregations, inclusion, seams, high spots, or hammer marks; blade crimped or warped.
103	Number of points per inch not within the specified tolerance.
104	Teeth not spring set, teeth not uniformly set or fail to provide ample clearance on either side of blade (types I, II, III, VII, X, XI and XIII).
105	Teeth not spring set, teeth not uniformly set, not adapted for both crosscutting and ripping, or teeth fail to provide ample clearance on either side of blade (types IV and VIII, style A).
106	Teeth not spring set, teeth not uniformly set, tooth set extends more than half the depth of the teeth, or teeth fail to provide ample clearance on either side of the blade (types V, VI and XIV).

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TABLE XVII.—*Classification of defects in accordance with Military Standard MIL–STD–105 (cont'd)*

Categories	Defects
107	Teeth not sharpened as specified; evidence of nicks, burrs, dull or incompletely formed teeth; teeth improperly formed or not even and of uniform height.
108	Reinforcing back not the required material, not secured to blade, fails to extend the full length of the blade, or (types I and X only) width and thickness of reinforcing back not as required (types I, VII and X).
109	One edge not toothed for ripping and the other for crosscutting (type II only).
110	Teeth not adapted for both crosscutting and ripping (compass and keyhole blades only) (type XI only).
111	Blade width not tapered (type III only).
112	Thickness of blade not within the specified maximum and minimum values; blade thickness not uniform within the specified tolerance (types I, II and III).
113	Thickness of cutting edge not within the specified maximum and minimum values; or blade thickness at cutting edge not uniform within the specified tolerance (types IV, V, VI, VIII, styles A and B, IX, XI, XII, XIII and XIV).
114	Blade width and thickness not tapered as required, or thickness of back at point within the specified maximum and minimum values (types IV, VIII, style A, XI and XII).
115	Blade width not tapered or handle not the required style (type XIII only).
116	Blade not taper ground or not uniformly tapered (types IX and XII only).
117	Tooth style (either edge) not as specified (type XIII, style B only).
118	Thickness of back at point not within the specified maximum and minimum values (types V, VI, IX and XIV).
119	Thickness of back at handle not within the specified maximum and minimum values (types V, IX and XIV only).
120	Blade not uniformly taper ground as specified (type VI only).
121	Blade not taper ground, not uniformly tapered from cutting edge to back and from handle to point, or amount of taper less than the specified minimum (types V and XIV only).
122	Blade not detachable, does not securely fit in the handle, or blade not tapered (type VIII, style B only).
123	Handle not pistol grip style, not metal, or handle fails to accommodate the specified width blades (type VIII, style B only).
124	Handle defective, not secured on blade as required, or handle not the required material; evidence of cracks, checking, soft spots, heavy knots, or wood not straight-grained (types I, II, III, IV, V, VII, VIII, style A, IX, X, XII, XIII and XIV).
125	Handle defective, fails to securely hold the saw blades, saw blades not readily interchangeable, or handle not the required material; evidence of cracks, checking, soft spots, heavy knots; wood not straight-grained.
126	Handle defective, or not secured on blade as required (type VI only).
Minor:	
201	Width under back not within the specified tolerance (types I, VII, and X).
202	Width of blade not within the specified tolerance (type II only).
203	Width of blade at point or at handle not within the specified tolerance (type III only).
204	Blade width at heel not within the specified tolerance (type VIII, style B).
205	Blade width at handle not within the specified maximum and minimum values (types IV and VIII, styles A and B).
206	Width of blade at butt or at point not within the specified tolerance (types V, VI, IX, XII and XIV).
207	Handle not smoothly finished, edges not rounded, or exposed surfaces not stained and coated as required.
208	Length of cutting edge not within the specified tolerance.
209	Handle not finished as required; evidence of fins or burrs (type VI only).
210	Blade not corrosion resistant or (if applicable) rust-resistant; finish incomplete or defective (type VIII, style B only).
211	Handle securing bolt and wing nut not corrosion resistant (types II and III).
212	Marking, manufacturer's name or trademark missing, illegible, or not permanent.

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4.5 Tests.

4.5.1 *Hardness.*—Each sample saw blade selected in accordance with 4.3.1 shall be subjected to a hardness test to determine conformance with the applicable requirement specified in section 3.

4.5.2 *Metal-cutting.*

4.5.2.1 *Type IX.*—Type IX metal-cutting hand saws shall be given a performance test as follows: They shall make a 24-inch cut in corrugated sheet metal freely and shall show no damage to the teeth after completion of cut.

4.5.2.2 *Type VIII, style B, type XI, and type XV, styles B and C.*—Type VIII, style B, type XI, and type XV, styles B and C saws shall be given the following test: They shall cut through a piece of 2- by 4-inch lumber and four 16-penny common wire nails driven into it at a 45-degree angle to the cut without damage to the blade.

4.5.3 *Bending test (types V and XIV).*—The blades of types V and XIV saws shall be subjected to the following test: The saw blade shall be bent slowly around a wooden mandrel 12 inches in diameter until the handle end of the saw is at 90 degrees with the point end; the bend shall be made half-way from the point to the handle. The saw shall be bent alternately through three cycles and shall show no greater deflection than ⅛ inch in the length of the saw blade. After the bending test has been made, the inspector shall procure a C clamp and a straight metal test bar at least 28 inches long. With the teeth of the saw lying adjacent to the edge of the test bar, the inspector shall clamp the saw blade to the test bar within one inch of the butt end of the blade, having the bowed side (if any) of the blade outwardly from the test bar. The inspector shall then measure the deflection at the point end of the saw blade, and if the deflection exceeds ⅛ inch, the saw shall be considered as nonconforming.

4.5.4 *Flammability test.*—The handle for test shall be clamped in a support with the handle extended outward in a horizontal position. Under the handle there shall be clamped a piece of 10-mesh Bunsen burner gauze about 5 inches square, in a horizontal position, ¼ inch below the bottom of the handle, and with about ½ inch of the handle extending beyond the edge of the gauze. A Bunsen burner with an efficient flame ½ to ¾ inch in height, shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the handle. At the end of 30 seconds the flame shall be removed and the sample allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the handle after the removal of the Bunsen burner. In case the handle does not continue to burn after the first ignition, the burner shall be replaced under the free end for a period of 30 seconds immediately following the extinction of the flame. If the handle does not continue to burn until the flame has reached the blade end of the handle after the second ignition, the sample shall be considered self-extinguishing. If the sample continues to burn, the burning rate shall not exceed 2 inches per minute.

4.5.5 *Possible test failures.*—Possible test failures are defined as follows:

(a) Hardness:
Hardness of blade not within the specified values on the Rockwell C scale.

(b) Nail-cutting test (nail-cutting blade only):
Nail-cutting blade not capable of cutting through the specified nails driven into a piece of lumber as required or evidence of dulled or damaged teeth after making the cut (type XI only).

(c) Nail-cutting test:
Saw not capable of cutting through the specified nails driven into a piece of lumber as required or

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evidence of dulled or damaged teeth after making the cut (type VIII, style B only).

(d) Metal-cutting:
Saw fails to make a clean-cut in the specified metal sheet or evidence of dulled or damaged teeth after making the cut (type IX only).

(e) Bending test (types V and XIV only):
Evidence of permanent set or cracks after blade has bent as specified.

5. PREPARATION FOR DELIVERY

(Civil agencies. — Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation and packaging.

5.1.1 *Level A.*—Saws shall be preserved and packaged in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Saws, cleaned, preserved, and packaged as specified shall be packed in accordance with level A of Military Specification MIL–P–15424.

5.2.2 *Level B.*—Saws, cleaned, preserved and packaged as specified shall be packed in accordance with level B of Military Specification MIL–P–15424.

5.2.3 *Level C.*—Saws, cleaned, preserved and packaged as specified shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Container shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type, style, class and length of saws required (see 1.2.1, 1.2.2 and applicable table of dimensions).
(*c*) Number of points per inch (see figure 1 and 3.19).
(*d*) Special requirements for cutting edge of type X saws (see 3.18).
(*e*) Levels of preservation, packaging and packing required (see section 5).
(*f*) Marking, if different (see 5.3).

6.2 Superseding data.—This specification supersedes Federal Specification GGG–S–65, dated December 8, 1954 and Federal Specification GGG–S–61a, dated November 3, 1942, in part. The indicated type of GGG–S–61a and GGG–S–65 are replaced by types of this specification as follows:

GGG–S–61a	GGG–S–65	GGG–S–65a
A	Type I	Type I
K	Type IV	Type IV
N	Type V	Type V
O	Type IX	Type IX
P	Type XIV	Type XIV
R	Type VIII, style A	Type VIII, style A
S	Type X	Type X
U	Type XI	Type XI

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